ISyE 6740 – Computational Data Analysis – Spring 2026
Project Proposal

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Title: Identifying Pit Stop Strategy Patterns in Formula One

Problem Statement

Across motorsports, strategy plays a vital role in determining race outcomes and is as important as driver skill and vehicle performance. In Formula One, entire teams are dedicated to deciding when to perform pit stops during a race while balancing tire wear, track position, and race pace. A poorly timed pit stop can cause a driver to lose positions, while a well-timed stop can provide a huge competitive advantage. Despite the importance of pit stop decisions, it is difficult to measure how effective different pit stop strategies are. However, the increasing availability of racing telemetry and historical race data makes it possible to analyze these strategies in greater detail.

The objective of this project is to analyze historical Formula One race data to examine how pit stop strategies influence race results. The study will examine the relationships between pit stop timing, the number of pit stops, and race outcomes such as finishing position or position change during a race. Machine learning techniques, including clustering and classification, will be used to identify patterns in pit stop strategies and evaluate how they affect race performance.

In summary, the project aims to answer the following question:

1. Can machine learning models identify relationships between pit stop timing, number of pit stops, and improvements in finishing position relative to starting grid position?
2. Are there distinct types of pit stop strategies that are found when clustering race data, and do certain strategies correlate with better race outcomes?

Data Source

The dataset used in this project is the "Formula 1 World Championship (1950 - 2024)" dataset found on Kaggle. It contains comprehensive historical race information covering Formula One seasons between 1950 and 2024. The dataset includes multiple relational tables for races, drivers, constructors, lap times, qualifying, and pit stops. Despite the dataset's large size, the project will focus only on the tables that are relevant for analyzing pit stop strategies.

The pit_stops.csv file contains detailed information about each pit stop during a race, including the driver, race, lap number, and duration of the pit stop. The lap_times.csv file provides lap-by-lap performance data for drivers, which can be used to evaluate changes in performance before and after pit stops. The results.csv file contains final race outcomes such as the finishing position and starting grid position, allowing comparisons between race strategy and the final results.

Additional tables such as races.csv, qualifying.csv, and drivers.csv will be used to provide contextual information about each race and driver. These tables can be merged using shared identifiers such as raceId and driverId to create a unified dataset suitable for analysis. The dataset spans around seventy years of Formula One racing which provides a strong sample size to ensure a strong analysis of pit stop strategies across different circuits, drivers, and seasons.

Methodology

Clustering Pit Stop Strategies

The first part of the analysis will identify common pit stop strategies using clustering techniques and group them into specific clusters. Each race instance will be represented by several numerical features from the dataset such as the total number of pit stops, the average pit stop duration, and the total time spent in pit stops during the race. These variables together describe how teams manage tire changes and track position during a race.

However, before clustering is applied, the dataset needs to be standardized so that all variables can be measured on a comparable scale. Using Principal Component Analysis (PCA), dimensionality reduction will be performed to transform the original variables into a smaller set of components that capture the large sources of variance in the data. This step can help reduce redundancy between variables and allows the relationships between pit stop strategies to be visualized.

After performing dimensionality reduction, the k-means clustering algorithm can be applied to group similar race strategies together. K-means partitions observations into k clusters by minimizing the within-cluster sum of squared distances between each observation and the centroid of its assigned cluster. Overall, the algorithm minimizes the objective function:

$$\sum_{i=1}^{k} \sum_{x \in C_i} || x - \mu_i ||^2$$

where $C_i$represents cluster $i$and $\mu_i$is the centroid of that cluster. This process groups races with similar pit stop characteristics into the same cluster.

Based on these features, several strategy clusters may emerge such as: early aggressive strategies (early pit stops with multiple stops), late conservation strategies (later pit stops with fewer stops), balanced strategies (moderate timing and duration), and high-time pit strategies where a long pit stop duration significant affected race performance. These clusters will represent different approaches teams use to balance tire management, race pace, and track position. Once clusters are formed, they can be compared with race outcomes such as finishing position and position improvement relative to starting grid position. The comparison helps determine which pit stop strategies associate with stronger race performance.

Predicting Position Improvement

The second stage of the analysis will apply classification models to examine how pit stop strategy influences changes in race position. A binary response can be created by comparing a driver's finishing position to their starting grid position. For example, if a driver finishes a race in a better position, that will be labeled as 1. Otherwise, it will be labeled as 0.

Each observation will include several features that describe race strategy and race performance. These features include the total number of pit stops, the lap of the first pit stop, total pit stop duration and starting grid position. Additional features can be created by combining information from the pit stop and lap time datasets. For instance, lap time data can be used to measure a driver's average pace before and after a pit stop. The difference between these values can help capture how tire changes or strategy timing affect performance during the race.

The first classification algorithm used will be logistic regression, which estimates the probability that a driver improves their race position based on the input features. Logistic regression is common for binary classification problems because it produces interpretable coefficients that show how each variable influences the predicted outcome (James et al., 2013).

The second model, Support Vector Machines, will classify observations by identifying a decision boundary that separates the two outcome groups. The algorithm will attempt to maximize the margin between observations that improved position and those that did not. This approach captures relationships that are more complex than those modeled by linear regression methods (Hastie et al., 2009).

Before training the models, the input variables will have to be standardized to ensure that differences in measurement do not affect the model's performance. Cross-validation can be used to evaluate the model's stability and reduce the risk of overfitting. Altogether, this modeling approach will allow the analysis to identify which pit stop strategy variables are most important for improvements in race position.

Evaluation Strategy

The performance of the two methodologies, clustering and classification models, can be evaluated using a few quantitative metrics. For the clustering stage, internal validation measures such as silhouette scores can be used to assess cluster quality. The silhouette score measures how similar each observation is to its own cluster compared to other clusters with higher values indicating a stronger separation between groups. Cluster results can also be visualized using lower-dimensional representations generated through the Principal Component Analysis and will help determine whether the clusters represent clear and interpretable strategy patterns.

The classification models will be evaluated using cross validation to estimate how well the model performs allowing several classification metrics to be calculated, including accuracy, precision, and F1-score. These metrics can measure how well the model predicts whether a driver improves their finishing position compared to their starting grid position.

The evaluation will also include an interpretation of the results in the context of Formula One race strategy. For example, if a cluster represents races with early pit stops and shorter pit stop durations, the average finishing position change for those observations can be calculated. If drivers within that cluster consistently gain positions compared to their starting grid position, the strategy may result in stronger race outcomes. By combining the statistical metrics with an interpretation of the results, the analysis will provide insight into how different pit stop strategies relate to race performance.

References

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